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Control of the phase evolution of kesterite by tuning of the selenium partial pressure for solar cells with 13.8% certified efficiency

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The control of the phase evolution during the selenization of kesterite $Cu_2ZnSn(S,Se)_4$ (CZTSSe) is crucial for efficient solar cells. Here, we regulate the phase-evolution kinetics of Ag-alloyed CZTSSe by applying a positive pressure in the reaction chamber at the initial stage of the annealing process. The partial pressure of Se decreases, reducing the collision probability between selenium molecules and the kesterite precursor during the initial formation of the crystals. This results in the precursor transforming into CZTSSe in a single step, without the formation of secondary phases. CZTSSe forms at relatively higher temperature than conventional methods, leading to high-crystallinity kesterite films with fewer defects. We demonstrate solar cells with a total area efficiency of 14.1% and a certified total area efficiency of 13.8%. This work provides insights into the selenization mechanism and phase evolution of kesterite absorbers, enabling efficient solar cells.

As one of the emerging solar cells, kesterite $Cu_2ZnSn(S_xSe_{1-x})_4$ (CZTSSe) solar cell has presented 13% of the power convention efficiency (PCE) based on environmentally friendly solution systems, which have shown attractive application prospects[1]. However, large open-circuit voltage deficit ($V_{OC,def}$) still restricts the further enhancement of the device performance, primarily due to complex phase composition and high deep-level defects in the CZTSSe absorber[2–4]. In fact, the high-temperature selenization process is an indispensable step to complete the phase evolution of the kesterite absorber; here, phase evolution is the phase transformation from CZTS to CZTSSe when the Se reacts with the precursor, which is mainly affected by Se partial pressure[5–8], reaction temperature and time[9,10], even precursor film composition and chemical environments[11,12] and so on. Besides, multi-element component CZTSSe has a narrow stable-phase region, which usually undergoes complicated phase-evolution pathways. Phase-evolution pathways are the ways to realize the phase evolution from CZTS to CZTSSe, and different paths imply dissimilar intermediate phases and different defect properties of the CZTSSe absorber[13–16]. Therefore, controlling CZTSSe phase-evolution pathways to avoid detrimental intermediate phases is the key to high-crystalline-quality CZTSSe absorbers with low defects and pure kesterite phase.

Selenization has been investigated for decades, including thermodynamic and kinetic processes and different reaction variables. Typically, understanding the resulting secondary phases is important to improve the phase evolution[3,13,17–20]. Currently, some groups have already explored the phase-evolution process and crystal growth of the CZTSSe absorber (or solution-based CIGS) during the selenization reaction from different precursor compositions[14–16], local chemical environment[12], element doping[21–25], atmosphere regulation[26–28], to chamber pressure controlling[17,29–32] and so on. These works contribute to

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understanding CZTSSe phase evolution and the reason for the $V_{OC,def}$ to some degree. That is, at the early heating-up stage of the selenization reaction, gaseous Se easily reacted with the CZTS precursor film to give intermediate phases, that is, binary phases (Cu_xSe, ZnSe, $SnSe_x$) and ternary phase (Cu_2SnSe_3, abbreviated as CTSe) with the temperature gradually increasing, depending on the composition of the precursor films derived from different preparation methods, for example, vacuum method, nanocrystalline method or solution method[13,17,20,27]. Besides, phase-evolution pathways can directly affect the formation of intermediate phases. For example, for Cu^+–Sn^{2+}-involved precursors, its phase-evolution process experienced successively from Cu_2S–ZnS–SnS to Cu_2Se–ZnSe–$SnSe_2$, then to CTSe–ZnSe and finally to CZTSe (ref. 15). Saucedo et al. introduced Ge to stabilize the Cu–Sn alloy, which phase-evolution process changed from metal stack to CTSe–ZnSe, then to CZTSe and simultaneously avoided binary Cu_xSe and $SnSe_x$ intermediate phases[14]. Xin et al. used Cu^+–Sn^{4+} as metal precursors in a DMSO (dimethyl sulfoxide) system, which is supposed to directly change from CZTS to final CZTSe, excluding all intermediate phases[15]. These above-mentioned works all focus on phase evolution and significantly reduce $V_{OC,def}$, especially Xin's work, which achieves a 13% efficiency kesterite solar cell, strengthening confidence in kesterite solar cells[1]. Different from the CZTSe, intermediate phases mainly originated from Se participation in reactions at relatively lower temperatures, and reaction rates are so fast (they could be completed within 2 min) (refs. 33,34). Obviously, the time window for regulating intermediate phases is too narrow. On the other hand, most high-efficiency CZTSSe devices were still based on a half-closed graphite box to complete the selenization reaction, which could increase the difficulty of reconciling those parameters including temperature and Se partial pressure[15,24,29]. Therefore, more effective regulation strategies are urgently required to avoid intermediate phases.

In this Article, we regulate the kinetic process of phase evolution by tailoring a positive chamber pressure to reduce Se partial pressure. The collision probability between the precursor and gaseous Se molecule can be decreased at the heating-up stage (200–400 °C) of the selenization reaction. Secondary phases resulting from decomposition on the surface and multi-step phase-evolution paths are thus inhibited significantly. Additionally, this strategy enables the phase evolution to start at a relatively higher temperature and thereby leading to a high-crystalline-quality kesterite absorber with fewer defects. The bulk defects are reduced by around one order of magnitude. Finally, we achieved a kesterite solar cell with 14.1% PCE (total area) and a certified 13.8% PCE (total area), which outperforms some prior reports[1].

Understanding the CZTSSe phase-evolution process

Selenization reaction of CZTS precursor films is the most important step among the fabrication processes of the CZTSSe solar cell, which covers phase evolution and crystal growth processes. Particularly, the phase-evolution process is crucial to the CZTSSe crystal quality and defect formations. As presented in Fig. 1a, the phase-evolution processes can be divided into three paths (Path I, Path II and Path III, details in Supplementary Note 1). For Path I and Path II, there are binary and ternary phases during phase evolution from precursor phase to CZTSSe phase, which will cause defects, according to previous research[4,14,15]. Thus, Path III in which precursor phase transformed directly to CZTSSe phase is desirable.

In this work, we prepared ACZTS precursor films based on a Ag-alloyed Cu^+–Sn^{4+}–MOE precursor solution (MOE: 2-methoxyethanol) in air ($ACZTS_{air}$), which is supposed to be more suitable for mass production. When this $ACZTS_{air}$ film was selenized under ambient atmospheric pressure, the occurrence of hetero-nucleation and intermediate phases was traced by interrupting at different time points. In this heating-up process, the $(Ag,Cu)_2ZnSn(S_xSe_{1-x})_4$ (ACZTSSe) was obtained with a Raman peak at 175 cm^{-1}, and other peaks varied from 215 cm^{-1} to 196 cm^{-1} as x changed from 0.5 to 0 while the temperature was increasing (Fig. 1b)[35] (Supplementary Note 2 provides details about Raman peak fitting). In the meantime, however, slight elemental Se (trigonal Se: 233 cm^{-1}, a-Se: 250 cm^{-1}) was left on the sample surface when the reaction was prematurely terminated (Supplementary Fig. 1)[36]. Typically, a Raman peak at 180 cm^{-1} assigned to the CTSe phase was also found when the selenization temperature reached 450 °C (ref. 18). The peak area increases as the temperature rises from 450 to 495 °C but disappears when further increasing the temperature to 535 °C (the wavenumber range of 150–390 cm^{-1} is presented in Supplementary Fig. 2). Compared with the precursor, the sample selenized at 350 °C for 300 s shows a weak peak at approximately 260 cm^{-1}, indicating the presence of Cu_xSe phase (Supplementary Fig. 3). It is thus suggested that the phase evolution undergoes Path I when heating up slowly. The formation of Cu_xSe and CTSe are mainly related to the precursor film, that is, oxygen–metal bonds are easily formed by thermally treating the ACZTS precursor film in the air, especially for alcohol-based (R-OH) solvent systems[37,38]. This can be further confirmed by the O 1*s* peak in Ar-etched X-ray photoelectron spectroscopy (XPS) (approximately 100 nm etching depth) by comparison of the MOE–air precursor and DMSO–N_2 precursor (Supplementary Fig. 4). The precursor film is generally considered to be poorly crystallized Ag–Cu–Zn–Sn–S along with a small amount of crystalline ACZTS, based on the X-ray diffraction (XRD) pattern and Raman spectra (Supplementary Fig. 5). In terms of thermodynamics, this precursor film could easily react with Se vapour during the selenization reaction, leading to some ACZTS decomposition accompanied with the appearance of binary or ternary phases, which may make the phase-evolution path complicated[20], even shifting from Path III to Path I or II.

In situ monitoring and tailoring kinetic phase evolution

Tailoring the chamber pressure is an effective way to obtain different Se partial pressure. According to the semi-empirical relationship proposed by Gilliland, assuming that binary mixed gases (N_2/Se(g)) are ideal gases, the diffusion coefficient of the gas molecules is proportional to p^{-1} (pressure) and $T^{3/2}$ (temperature), based on molecular dynamics theory[39]. The variation tendency between Se partial pressure and chamber pressure is also theoretically simulated (details in Methods, Supplementary Note 3, Supplementary Fig. 6 and Supplementary Table 1). The results show that the first 100 s is the most dramatic stage of Se partial pressure (Supplementary Fig. 7), and positive chamber pressure can decrease the gaseous Se diffusion rate to a certain extent, which leads to the reduction of Se partial pressure (Supplementary Fig. 8). To verify this experimentally, the Se partial pressure in the selenization reaction was in situ monitored by our lab-made selenization furnace (detailed measurement process is given in Supplementary Note 4 and Supplementary Fig. 9). Variation tendencies of Se partial pressure with the reaction time under different positive pressures are presented in Fig. 1c, and all the curves are divided into three parts corresponding to heating up, equilibrium and naturally cooling stages in the whole selenization process. With the chamber pressure increasing, the Se partial pressure rising rate is reduced at the heating-up stage, but the time to reach the stable Se vapour value is distinctly prolonged, well consistent with the above simulation (different chamber pressure can be realized by increasing or decreasing the partial pressure of N_2 gas) (Fig. 1d). Therefore, regulating the chamber pressure mainly affects the early stage in the selenization reaction.

On the basis of CZTSSe absorbers under different chamber pressures, we further fabricated kesterite solar cells with a configuration of Mo/ACZTSSe/CdS/ZnO/ITO/Ni-Al/MgF_2. According to statistical results of photovoltaic parameters (all based on total area) in Fig. 1e, with the chamber pressure increasing, both FF and V_{OC} are remarkably improved while the J_{SC} is slightly decreased mainly due to different band gap by the Se/(S + Se) ratio (X-ray fluorescence data in Supplementary Fig. 10), and

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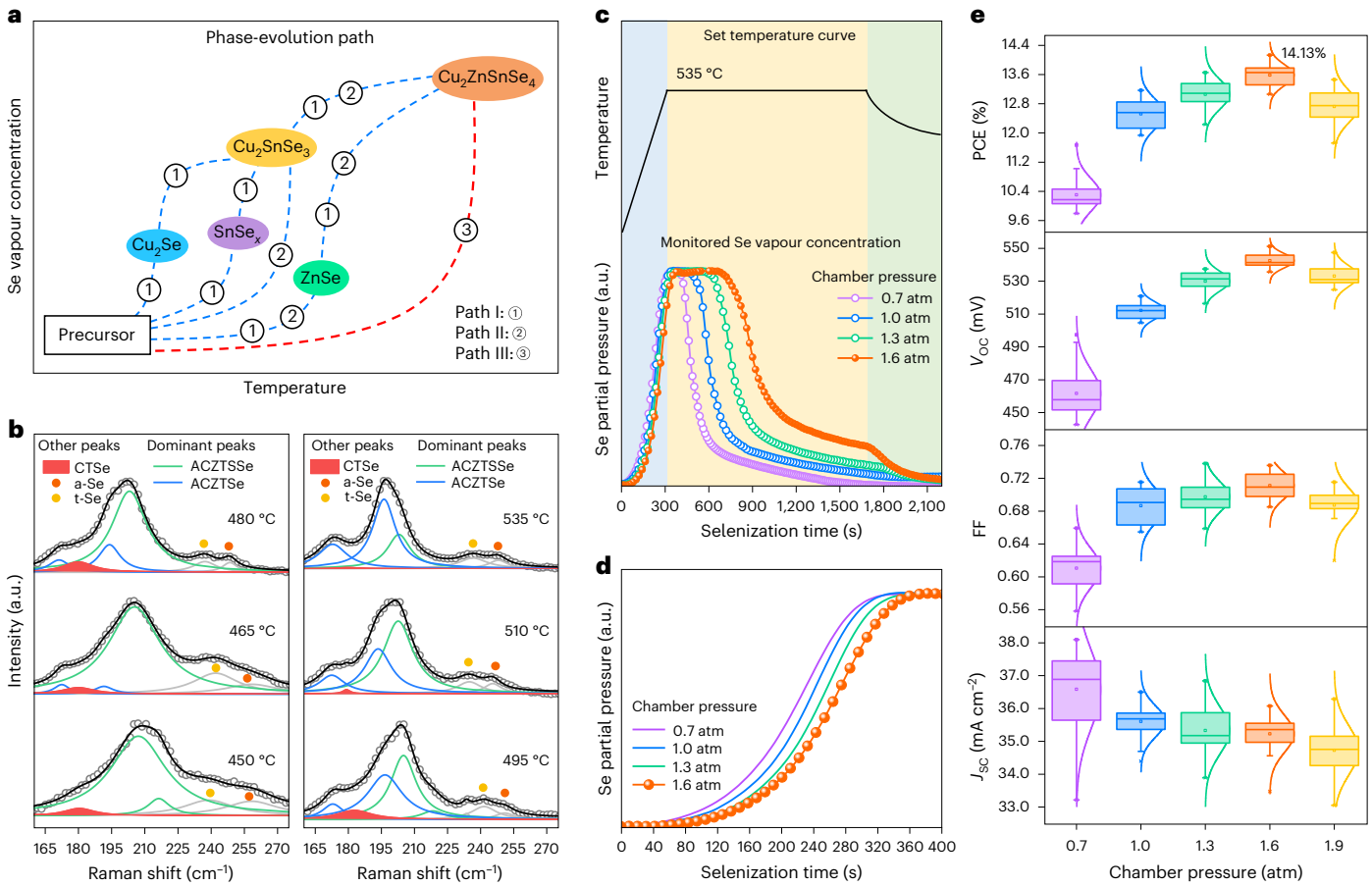


Fig. 1 | The method to regulate the kinetic process of phase evolution and the resulting solar cell performance. a, Schematic diagram of the phase-evolution path (normalized Se partial pressure vs temperature) (detailed reaction equations of Path I, II, III in Supplementary Note 1). **b**, Raman spectra (532 nm excitation wavelength) of samples obtained at 450, 465, 480, 495, 510 and 535 °C, with a relatively slow heating-up rate of 0.63 °C s^{-1} under ambient pressure. t-Se, trigonal selenium; a-Se, amorphous selenium. Black solid lines on top of raw data are the fitted curves using Lorentzian fitting. **c**, Variation trend of the temperature and Se partial pressure vs selenization time under various chamber pressures derived from real-time monitoring results. Blue shading indicates the heating-up stage, yellow indicates the equilibrium stage and green indicates the naturally cooling stage. **d**, Enlarged view of the heating-up stage in Fig. 1c. **e**, Statistical results of photovoltaic parameters (with anti-reflection coating): PCE, V_{OC}, fill factor (FF) and short-circuit current density (J_{SC}). The total area of kesterite solar cells is 0.28 cm^2 in our lab. The box plot denotes median (centre line), 25th (bottom edge of the box), 75th (top edge of the box), 150th (upper whisker) and −50th (lower whisker) percentiles. Each box contains 24 solar cells.

the average PCE reaches to the highest value of 13.59% under 1.6 atm. This variation tendency reflects the positive chamber pressure could bring better cell performance of kesterite solar cells. Further investigation reveals that 1.6 atm chamber pressure postpones the crystallization process to approximately 410 °C (Supplementary Fig. 11), which can avoid the formation of those intermediate phases with relatively low Gibbs free energy in the temperature range of 200 to approximately 400 °C. Besides the MOE solvent system, our positive-pressure selenization strategy is also suitable for other precursor systems, such as DMF (N,N-dimethylformamide) and DMSO (Supplementary Figs. 12 and 13). The highest total-area PCEs are 13.19% for the DMF precursor (1.6 atm) and 12.89% for the DMSO precursor (1.3 atm), respectively (Supplementary Fig. 14).

Characteristics of phase evolution and crystal growth

Influence of different positive pressures on the early phase evolution of kesterite has been explored first. Under different chamber pressures, the $ACZTS_{air}$ film was selenized by directly raising the temperature to 535 °C in 60 s, then holding at 535 °C for 1,300 s and finally naturally cooling to room temperature. On the basis of the above discussion (Fig. 1b), the rapid heating-up rate (60 s to 535 °C) was set to avoid intermediate phases; here we focus on the reaction stage at 535 °C. Energy dispersive spectrometer (EDS) element maps show that for the $ACZTS_{air}$ precursor film, Zn is slightly enriched on the surface while the distributions of Ag, Cu and Sn are basically uniform (Fig. 2a (left) and Supplementary Fig. 15). However, for samples selenized under ambient pressure (Fig. 2a (middle) and Supplementary Fig. 15), the diffusion rate of Cu from film bottom to surface is higher than that of Sn, thus leading to a Cu–Zn-rich and Sn-depleted surface. It is supposed that under ambient pressure, the reaction between ACZTS and Se vapour at the film surface may cause decomposition of ACZTSSe and the Cu_xSe simultaneously forms in the initial heating-up stage. Therefore, the Cu_xSe formation process will promote the Cu diffusion to the surface. These two processes mutually reinforce each other, resulting in the above difference. In this way, the phase evolution in the entire film shows a characteristic of one-step transition but with a small number of intermediate phases on the surface, that is, most Ag–Cu–Zn–Sn–S in the bulk precursor reacts with Se to directly transform to the ACZTSSe phase, but a little Ag–Cu–Zn–Sn–S on the surface decomposes and transforms from Cu_xSe to CTSe and finally to ACZTSSe phase. For positive-pressure samples (1.3, 1.6 and 1.9 atm), due to much lower Se partial pressure in the early stage, the thermodynamic condition for forming Cu_xSe cannot be reached; therefore, the entire film is directly

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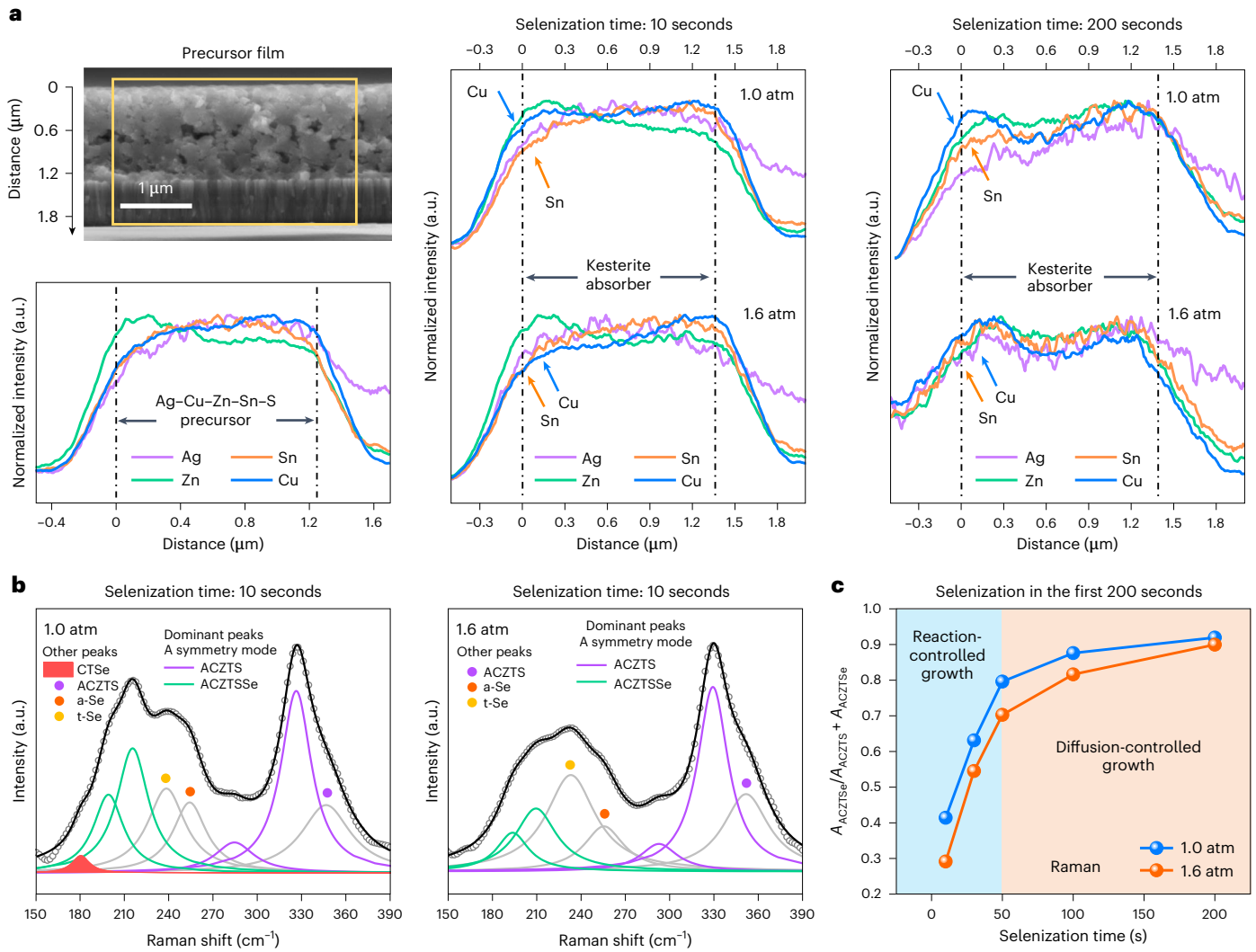


Fig. 2 | Influence of chamber pressure on the kinetic process of phase evolution. a–**c**, The samples obtained under various conditions are denoted as *p*-535-*t* (*p*: pressure (atm); *t*: selenization time (s)). **a**, EDS element maps (normalized by the maximum value) of the Ag–Cu–Zn–Sn–S precursor (left), 1.0-535-10 (middle) and 1.6-535-10 (right). The mapping zone is marked with a rectangle in the image. **b**, Raman spectra (532 nm excitation wavelength) of 1.0-535-10 (left) and 1.6-535-10 (middle). Black solid lines on top of raw data are the fitted curves using Lorentzian fitting. **c**, R_{Raman} of *p*-535-*t* obtained by Raman spectra vs selenization time in the early 200 s, $p = 1.0$ and 1.6, $t = 10, 30, 50, 100$ and 200. R_{Raman} is the peak area ratio of the ACZTSe phase to the sum of ACZTS and ACZTSe phases. $R_{Raman} = A_{ACZTSe}/(A_{ACZTS} + A_{ACZTSe})$. *A*: Raman peak area of the ACZTS phase (290–350 cm^{-1}) or ACZTSe phase (160–220 cm^{-1}).

transformed from Ag–Cu–Zn–Sn–S to ACZTSSe, with the formation of a secondary phase well avoided (Fig. 2a (right) and Supplementary Fig. 16).

Raman spectra further confirm the above conclusion (Fig. 2b and Supplementary Fig. 17), that is, the CTSe peak is detected in 1.0–535–10 s samples but cannot be detected in positive-pressure samples (for example, 1.6 atm) (Fig. 2a (middle), Fig. 2b (right) and Supplementary Fig. 18). Obviously, positive chamber pressures in the selenization process can efficiently retard the formation of those thermodynamic priority intermediate phases (that is, CTSe), also confirmed by XRD patterns (Supplementary Fig. 19). The effect of positive pressures on the early phase-evolution process (within early 200 s) mainly lies in the selenization reaction degree, which is evaluated by a semi-quantitative method: the R_{Raman} value ($R_{Raman} = A_{ACZTSe}/(A_{ACZTS} + A_{ACZTSe})$) (Fig. 2c). The less the R_{Raman} value is, the slower the conversion from amorphous ACZTS to crystallized ACZTSSe, and thus the slower the reaction rate is. It can be seen from Fig. 2c that in the early 200 s reaction stage, the R_{Raman} under positive chamber pressures is lower than that under ambient pressure but can reach a similar value at 200 s. It is thus proposed that the starting time of the phase evolution has been delayed under positive pressures, and the difference in phase evolution is mainly reflected in the very earlier reaction stage. It is worth noting that the R_{Raman} versus selenization time exhibits characteristic double exponential growth, also suggesting a rapid growth in the early 50 s (blue area), followed by a slow growth process (orange area). This is consistent with the crystal growth theory proposed in early work, namely, the crystal growth is involved in two stages: one is a reaction-controlled growth stage with obvious crystal-orientation growth, and the other is a diffusion-controlled growth stage with round grains on the surface[20].

In our work, the reaction-controlled growth stage lasts only about 50 s, while the subsequent diffusion-controlled growth stage will last until ending. In fact, the most critical phase-evolution process is determined by the reaction-controlled growth stage; therefore, how to regulate this reaction-controlled growth stage becomes crucial. On the basis of the above discussion, we reach a conclusion that increasing chamber pressure can inhibit the earlier reaction between Se vapour and the precursor, and the early phase-evolution stage is delayed around 30 s to 50 s; therefore, intermediate phases and Cu, Sn heterogeneous

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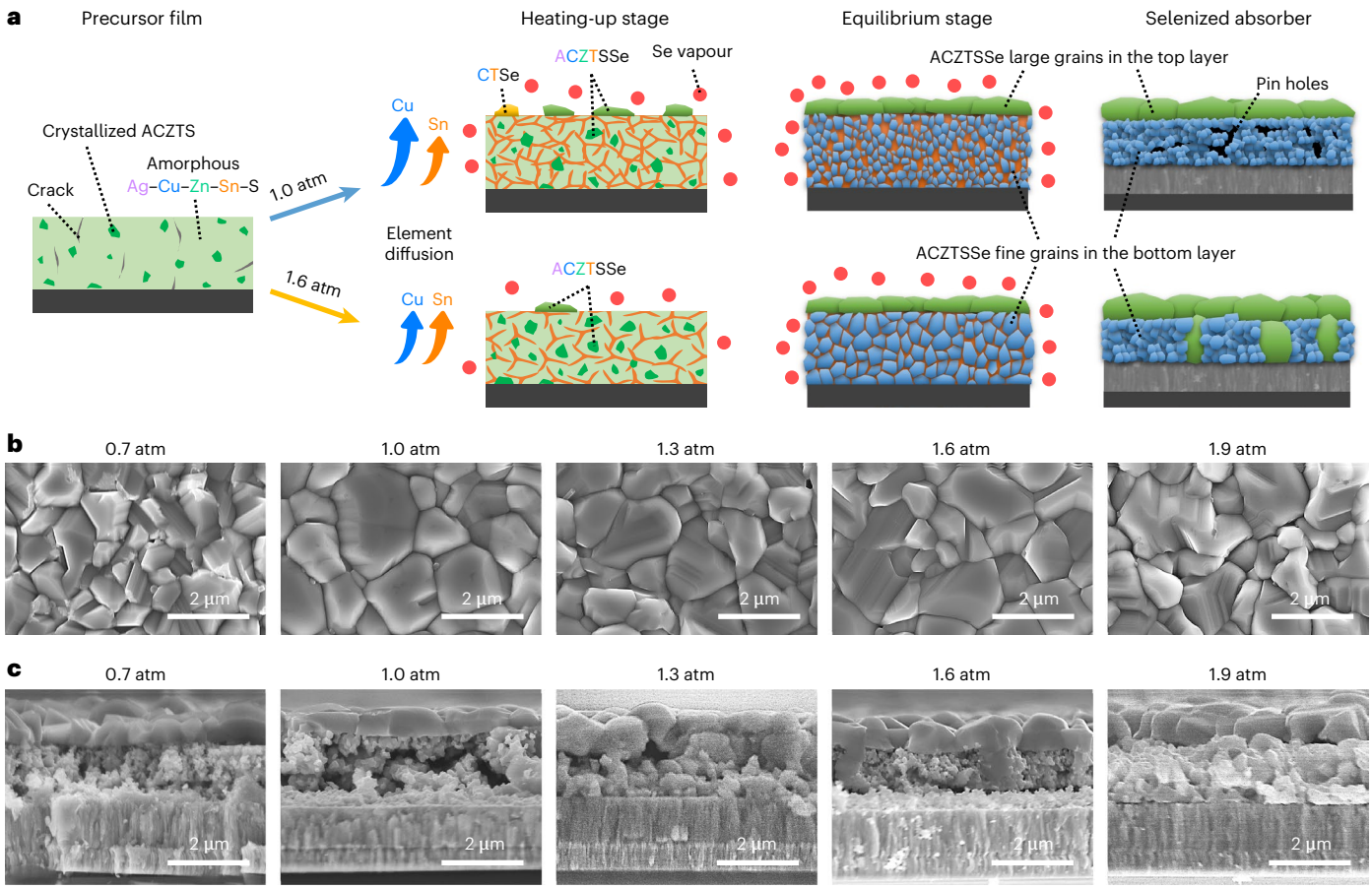


Fig. 3 | Influence of chamber pressure on kesterite morphology. a, Schematic diagram for the selenization process. Green and blue grains in the equilibrium stage and final absorber represent ACZTSSe large grains and ACZTSSe fine grains, respectively. **b**, Top-view scanning electron microscopy (SEM) images of the final ACZTSSe absorbers under 0.7, 1.0, 1.3, 1.6 and 1.9 atm chamber pressures. **c**, Cross-section SEM images of final ACZTSSe absorbers under 0.7, 1.0, 1.3, 1.6 and 1.9 atm chamber pressures.

diffusions will be effectively avoided, and the kinetic regulation window of the ACZTSSe phase can also be widened. This strategy is beneficial for regulating the overall phase-evolution pathway.

A schematic diagram is further suggested about the crystal growth mechanism in the selenization process by regulating positive chamber pressures (Fig. 3a). For element diffusion and phase evolution on the surface, it is consistent with the above discussion. For crystal growth, however, sufficient Se vapour uniformly penetrates into the precursor film with a loose, porous surface, then crystallization occurs preferentially in the top layer with island-like grains. Small grains are finally mixed and ripen to allow large grains (Fig. 3b and Supplementary Fig. 20). In fact, nucleation and crystallization occur simultaneously on the surface and inside the precursor film; however, fine grains are formed in the bottom layer rather than large grains in the top layer, which may bring more grain boundaries and bulk defects that affect device efficiency[40]. When the chamber pressure is increased, some fine grains in the bottom can fuse to larger grains with fewer pores (Fig. 3c and Supplementary Fig. 21). When 1.6 atm of the chamber pressure is adopted, relatively larger grains form in the bottom layer, and better contact with the Mo substrate is obtained (Fig. 3c). This dense and fewer-pin holes morphology in the bottom layer can reduce grain boundaries and resulting defects and improve the back-interface property.

Surface defect properties and device performances

Effect of positive chamber pressures on surface states and defect properties of final ACZTSSe absorbers, 1.0-ACZTSSe and 1.6-ACZTSSe (chamber pressure: 1.0 and 1.6 atm), are explored. The potential profiling demonstrates that positive chamber pressure results in a downshift in the Fermi level of the ACZTSSe film and relatively higher shallow acceptor carriers and approximately 50 mV lower potential value of the 1.6-ACZTSSe absorber than that of the 1.0-ACZTSSe absorber (Fig. 4a,b). On the other hand, a positive chamber pressure can also bring much more uniform potential of grain surface and grain boundary, indicating both impurity phases and defects at the grain boundary are significantly reduced for all the positive-pressure samples (Supplementary Figs. 22 and 23). Moreover, based on previous work of perovskite solar cells[41], the relatively lower difference between relative potential values in the dark and under illumination means fewer surface states on the *p*-type ACZTSSe surface (Fig. 4c and detailed discussion in Supplementary Note 5). Obviously, under the positive chamber pressure, the crystallinity of the ACZTSSe absorber has been significantly improved.

Surface defect property of as-obtained ACZTSSe is also investigated by Raman spectra (325 nm excitation wavelength), which usually can detect approximately 30 nm depth from the film surface. Peak area ratios denoted as $A_{168}/(A_{168}+A_{190})$ and $A_{253}/(A_{253}+A_{190})$, represent point-defect clusters of A type ($V_{Cu}+Zn_{Cu}$) and B type ($Zn_{Sn}+2Zn_{Cu}$), respectively, where characteristic ACZTSSe phase (190 cm^{-1}) and Cu-, Zn- and Sn-related defects (168, 230 and 253 cm^{-1}) are used (Fig. 4d and Supplementary Fig. 24) (ref. 42). The ACZTSSe absorber derived from positive chamber pressures has relatively higher V_{Cu} concentration due to lower $A_{168}/(A_{168}+A_{190})$ (1.0-ACZTSSe: 0.5395, 1.3-ACZTSSe: 0.6073, 1.6-ACZTSSe: 0.3454, 1.9-ACZTSSe: 0.3138; detailed discussion in Supplementary Note 6). Moreover, deep-level defect Sn_{Zn} concentration

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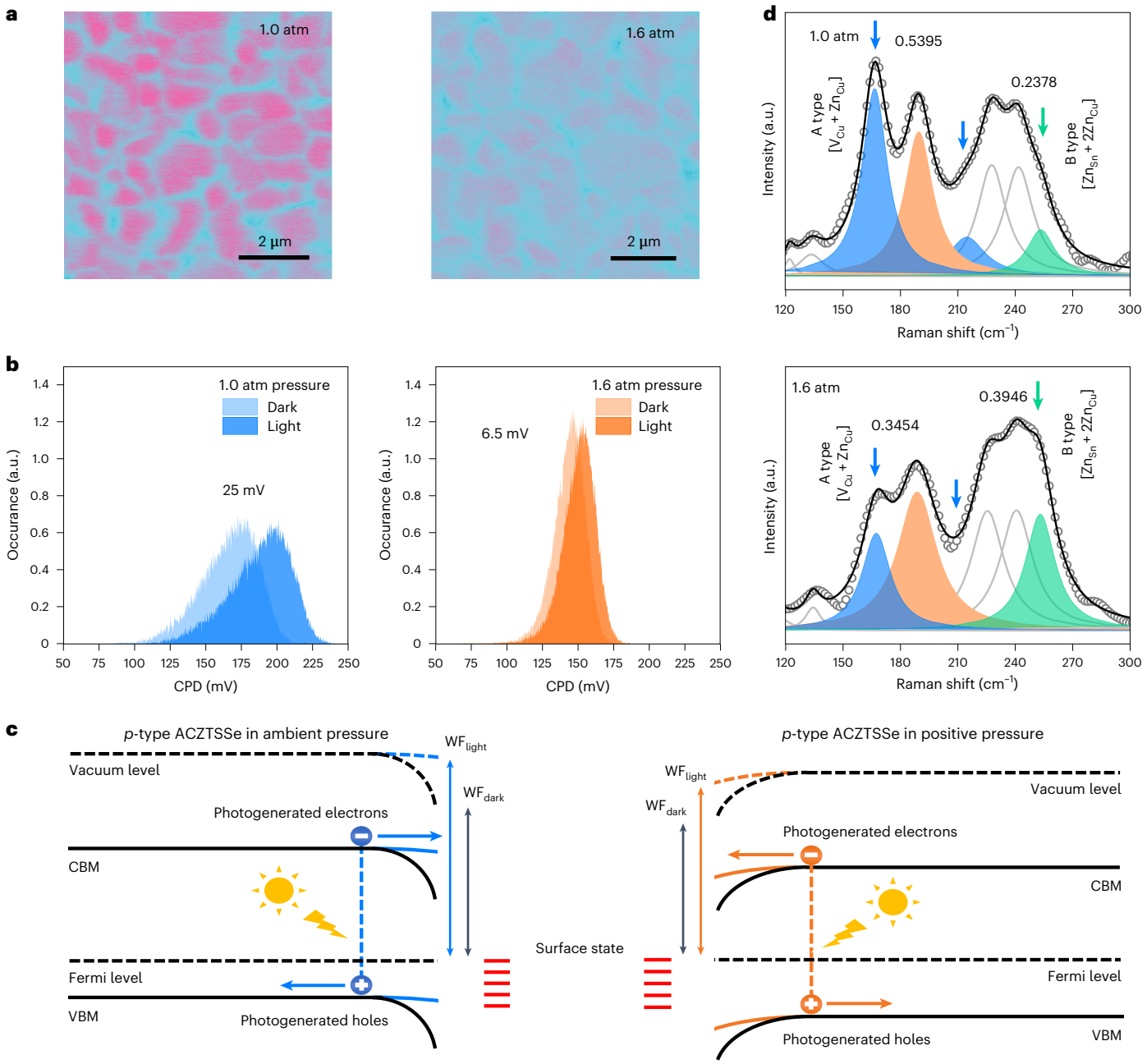


Fig. 4 | Surface state and defect properties. a, KPFM images of 1.0-ACZTSSe (left) and 1.6-ACZTSSe (right). **b**, Contact potential difference (CPD) distribution of 1.0-ACZTSSe (left) and 1.6-ACZTSSe (right) in dark and under illumination. **c**, Schematic energy band diagram for *p*-type ACZTSSe surface under ambient and positive pressure (CBM, conduction band minimum; VBM, valence band maximum; WF_{dark}, work function of ACZTSSe in dark; WF_{light}, work function of ACZTSSe under illumination). The solid lines and dotted lines represent dark and light conditions, respectively. **d**, Raman spectra (325 nm excitation wavelength) of final kesterite absorbers: 1.0-ACZTSSe (top) and 1.6-ACZTSSe (bottom). Black solid lines on top of raw data are the fitted curves using Lorentzian fitting.

in the 1.6-ACZTSSe is also decreased due to the obvious increase of the $A_{253}/(A_{253}+A_{190})$ (1.0-ACZTSSe: 0.2378, 1.3-ACZTSSe: 0.3398, 1.6-ACZTSSe: 0.3946, 1.9-ACZTSSe: 0.2565). Therefore, less deep-level defect is obtained for the 1.6-ACZTSSe absorber and thus is beneficial for reducing carrier recombination and V_{OC} deficit to further improve device efficiency of kesterite solar cells.

The device based on the 1.6-ACZTSSe absorber (1.6-device) presents 35.74 mA cm^{-2} of J_{SC}, 551.20 mV of V_{OC}, 71.73% of FF, yielding the best PCE (Fig. 5a) at 14.13% and a certified 13.8% PCE (Supplementary Fig. 25). The integrated J_{SC} from external quantum efficiency (EQE) spectra is 37.1 mA cm^{-2} (without shield of grid lines), in good agreement with the active-area J_{SC} (37.1 mA cm^{-2}) (Fig. 5b). Its band-gap energy (E_g) is 1.097 eV, estimated from the first derivative of the EQE curve in long wavelength; moreover, the band gap of ACZTSSe layers gradually widens as the chamber pressure increases (Supplementary Fig. 26 and Supplementary Table 2). The V_{OC} deficit can be reduced to 0.3042 eV ($V_{OC}^{SQ} - V_{OC}$). This result is much better than the record kesterite cells (Fig. 5c) and has especially better carrier transportation property (less recombination and better transportation)[12,26,43–45]. As it is usually hard to simultaneously obtain high V_{OC} and J_{SC} in kesterite solar cells, the parameter $v \times j$ can be used to reflect the potential for absorber materials. For our devices, the parameter $v \times j$ is above 50% of $V_{SQ} \times J_{SQ}$;

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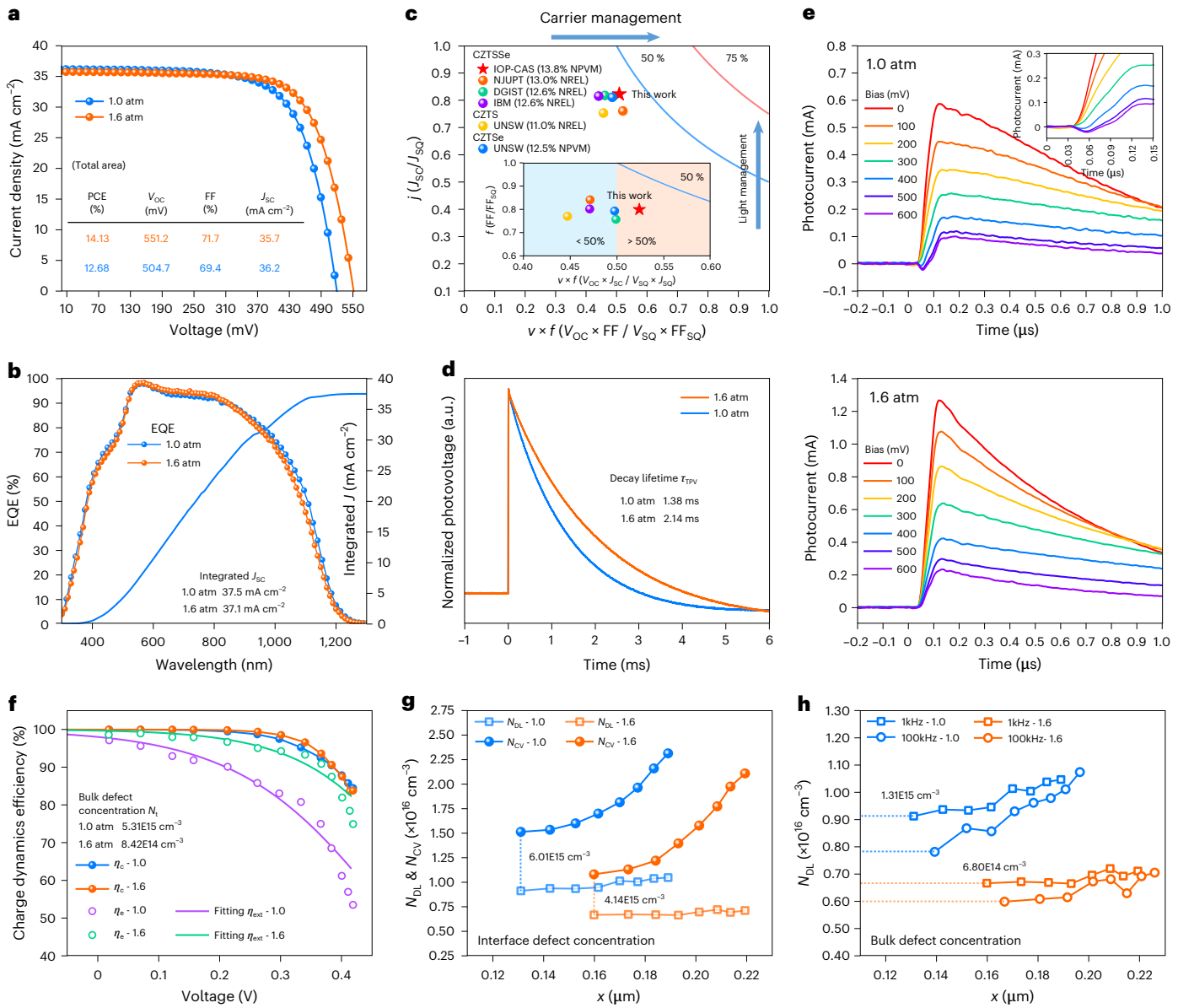


Fig. 5 | Performance of photovoltaic devices. a, J–V curves of p-device (p = 1.0 and 1.6 atm). **b**, EQE curves and integrated J_{SC} of the best devices (1.0 and 1.6 atm). The measuring active area is around 0.26 cm^2 (without the grid line). **c**, Fraction of Shockley–Queisser (S–Q) detailed balance limit for V_{OC},J_{SC} and FF achieved by record kesterite cells and our result[12,26,43–45] (IOP-CAS, Institute of Physics, Chinese Academy of Sciences; NJUPT, Nanjing University of Posts and Telecommunications; DGIST, Daegu Gyeongbuk Institute of Science and Technology; IBM, International Business Machines; UNSW, University of New South Wales; NREL, National Renewable Energy Laboratory; NPVM, National PV Industry Measurement and Testing Center). The parameter $v \times f((V_{OC} \times FF)/(V_{SQ} \times FF_{SQ}))$ represents carrier management, and j (J_{SC}/J_{SQ}) represents light management. The inset image also reflects S–Q limit information, but with $v \times j$ ($(V_{OC} \times J_{SC})/(V_{SQ} \times J_{SQ})$) as horizontal axis and f (FF/FF$_{SQ}$) as vertical axis. **d**, Transient photovoltage (TPV) spectra of p-device (p = 1.0 and 1.6 atm). τ_{TPV}: decay lifetime of TPV curve. **e**, Transient photocurrent (TPC) spectra of 1.0-device (top) and 1.6-device (bottom) under various forward biases. **f**, Modulated TPV/TPC spectra of p-device (p = 1.0 and 1.6 atm) (N_t, bulk defect state density; η_C, charge collection efficiency; η_E, charge extraction efficiency; Fitting η_{ext}, fitting curve of η_E). Purple and green solid lines on top of raw data are the fitted curves, and the fitting details are given in Supplementary Note 7. **g**, Carrier concentrations obtained by driven level capacity profiles (N_{DL}) and capacitance-voltage (N_{CV}) of p-device (p = 1.0 and 1.6 atm) under 1 kHz. X, depletion width. **h**, N_{DL} of p-device (p = 1.0 and 1.6 atm) under 1 and 100 kHz.

however, for the other devices, this parameter is usually below 50%, suggesting a better prospect for this kinetic regulation strategy. The highly efficient kesterite solar cell also shows an excellent long-term stability in ambient environment without encapsulation (three months; Supplementary Table 3).

Comparison of transient photovoltage (TPV) spectra reveals that carrier recombination loss in the 1.6-device has been reduced and a longer τ_{TPV} (decay lifetime of TPV curve) is obtained (1.38 vs 2.14 ms under 0 V bias) (Fig. 5d). For transient photocurrent (TPC) spectra (Fig. 5e), however, the 1.0-device is an exception because negative photocurrent signals appear for about 0.03 μs before rapid rising when >400 mV forward bias is applied, unlike for the 1.6-device. This further confirms much better back-interface contact property of the 1.6-device than that of the 1.0-device[24]. Modulated TPV/TPC spectra also demonstrate that the charge extraction efficiency (η_E) has been significantly improved as chamber pressure varies from the ambient to positive pressure, consistent with the fact that the defects of the bulk CZTSSe absorber are reduced and recombination is suppressed[46–48].

The bulk defect state densities (N_t) are estimated to be 5.31×10^{15} and 8.42×10^{14} cm^{-3} for the 1.0-device and 1.6-device, respectively (Fig. 5f and detailed discussion in Supplementary Note 7), basically consistent with their surface properties (Fig. 4). And relatively lower E_a (photoluminescence quenching activation energy) and E_A (activation energy of dominated recombination process from temperature-dependent J–V measurements) also verify better crystallinity of ACZTSSe films and device properties, respectively (Supplementary Note 8 and Supplementary Fig. 27).

Figure 5g,h presents interface defects and bulk defects derived from carrier concentrations by driven level capacity profiles (N_{DL}) and capacitance-voltage measurement (N_{CV}) under different frequencies[49]. Interface defect concentrations of the 1.0-device and 1.6-device are 6.01×10^{15} and 4.14×10^{15} cm^{-3}, respectively, which are derived from the N_{DL} and N_{CV} in low frequency (1 kHz). Bulk defect concentrations of the 1.0-device and 1.6-device are 1.31×10^{15} and 6.80×10^{14} cm^{-3}, respectively, which can be described by the difference between N_{DL} in low frequency (1 kHz) and N_{DL} in high frequency (100 kHz) (detailed parameters of electrical properties in Supplementary Table 4). Obviously, positive pressures indeed help to reduce bulk defect concentration by about one order of magnitude and interfacial defects, finally leading to high-quality CZTSSe absorbers and high-performance photovoltaic devices.

Conclusions

We have regulated the phase-evolution process by collaboratively tailoring the early selenization reaction rate in a half-closed graphite box. By increasing positive chamber pressure, Se partial pressure has been intentionally decreased, which can also reduce the collision probability between ACZTS precursor film and gaseous Se molecules during the heating-up stage (200–400 °C) of selenization. Starting from regulating the kinetic process of the reaction, the decomposed secondary phases (Cu_xSe) on the surface and resulting multi-step phase-evolution paths are thus inhibited significantly. This strategy enables the phase evolution to start at a relatively higher temperature and thereby obtain a high-crystalline-quality ACZTSSe absorber with fewer defects on the surface. The surface states are reduced, and the bulk defect density is reduced by around one order of magnitude (from 5.31×10^{15} to 8.42×10^{14} cm^{-3}). Finally, we achieved a kesterite solar cell with a self-measured PCE of 14.1% (total area) and a certified PCE of 13.8% (total area). This work provides a kinetic regulation strategy for further understanding and regulating the phase-evolution process of kesterite, especially optimizing the phase-evolution path to achieve highly efficient kesterite solar cells.

Methods

Materials

CuCl (99.999%, Alfa), $Zn(CH_3COO)_2$ (99.99%, Aladdin), $SnCl_4$ (99.998%, Macklin), AgCl (99.5%, Innochem), thiourea (99.99%, Aladdin, recrystallized twice), 2-methoxyethanol (MOE, 99.8%, Aladdin), dimethyl sulfoxide (DMSO, 98%, Alfa Aesar), dimethylformamide (DMF, 98%, Alfa Aesar), Se pellets (99.999%, Zhong Nuo Advanced Material), $CdSO_4 \cdot 8/3H_2O$ (99.99%, Aladdin), ammonium chloride (≥99.5%, Sinopharm Chemical Reagent Co. Ltd.) and ammonium hydroxide (25.0–28.0%, Sinopharm Chemical Reagent Co. Ltd.) were used to prepare the precursor solution. The chemicals were used directly without further purification.

Film preparation

The precursor solution was prepared as followed: AgCl, CuCl, $Zn(CH_3COO)_2$, $SnCl_4$ and thiourea were dissolved in the solvent (MOE, DMSO, DMF) and stirred at 60 °C for an hour to obtain a colourless solution, then AgCl was dissolved in the solution for another an hour to get a final colourless solution (MOE–air precursor). The molar ratio of Ag / (Ag + Cu), (Ag + Cu) / (Zn + Sn), Zn/Sn and thiourea/metal are 0.1, 0.75, 1.12 and 1.7, respectively, and the concentrations of metal elements and thiourea are 1.88 mol l^{-1} and 3.20 mol l^{-1}, respectively. The ACZTSSe precursor films were prepared by spin coating the precursor solution on cleaned Mo substrates at a spin speed of 3,000 r.p.m. s^{-1} for 25 s, followed by annealing on a 280 °C hot plate in the air. For the DMSO (DMF) solvent, we spin coated the precursor solution at a spin speed of 2,500 r.p.m. s^{-1} (3,500 r.p.m. s^{-1}) for 25 s, followed by annealing on a 360 °C hot plate in the N_2-filled glovebox. The coating and annealing processes were repeated several times for the desired 1.2–1.5 μm thickness precursor films. Then, as-prepared precursor films were annealed with N_2 flow in a selenium-contained graphite box, which was put inside the selenization furnace fabricated in our lab. For comparison, Xin's work was also repeated using the same composition in DMSO (DMSO-N_2-precursor).

Device preparation

A device structure with Mo/ACZTSSe/CdS/ZnO/ITO/Ni/Al/MgF_2 was fabricated successively by a ACZTSSe layer derived from selenization towards the spin-coated precursor film, chemical bath-deposited CdS buffer layer, radio frequency magnetron-sputtered i-ZnO combined with indium tin oxide (ITO) window layers and thermally evaporated Ni/Al collection grid and MgF_2 anti-reflection layer. The total area of kesterite solar cells is defined to be 0.28 cm^2 in our lab measurement. The total area of the certified champion kesterite solar cell is 0.2668 cm^2, which is certified by the National PV Industry Measurement and Testing Center.

Simulation

We simulate the effect of chamber pressures on the evaporation of Se droplets in a high-temperature environment (details in Supplementary Note 3). The selenium droplet was placed in a closed space (graphite box), and the initial temperature of the simulation was set to be 535 °C (actual reaction temperature). The simulation does not involve the heating-up process.

Geometric dimensions

Because the graphite box (containing selenium droplets and samples) we used has rotational symmetry, three-dimensional (3D) information can be reflected by a radial two-dimensional (2D) cross section (Supplementary Fig. 6a). To simplify the model, we used the cross section (Supplementary Fig. 6b) for simulation and consider only the effect of Se vapour generated by one Se droplet on the sample. The dimensions of the 2D rectangle are 35×7 mm^2. An Se droplet with a radius of 1.5 mm is placed on the left side of the 2D model (centre of the 3D model). The sample (20×2 mm^2) is in the centre of the rectangle.

Meshing

The meshing of the 2D model is shown in Supplementary Fig. 6c using free triangular mesh. Dynamic mesh is used in the edge of the Se droplet (the interface between liquid Se and gaseous Se) to simulate the shrinkage of the Se droplet caused by gas volatilization in the actual reaction process.

Parameter list

The parameters used for simulation are in Supplementary Table 1.

Characterization

Optical images of samples were measured on a metallographic microscope (OLYMPUS, BX61). XPS measurement was carried out on an ESCALAB 250Xi (Thermo Fisher) instrument, and ≈100 nm depth XPS was obtained by argon etching. Raman spectra were collected by a Raman spectrometer (Lab-RAM HR Evolution, HORIBA) by using a 532 nm or 325 nm excitation laser with the power of 6 mW. The film compositions were determined by an energy dispersive X-ray fluorescence spectrometer (EDX-7000, Shimadzu). Element maps were obtained by energy dispersive spectrometer (AZtec X-Max 50). Scanning

Electron microscopy (SEM) images were taken by a scanning electron microscope (Sigma 300) under 10 kV at various magnifications. X-ray diffraction (XRD) patterns were obtained on an X-ray diffractometer with Cu Ka as the radiation source (Empyrean, PANaltcal). Atomic force microscope (AFM) and Kelvin probe force microscope (KPFM) images were obtained on a Bruker Multimode 9. Steady-state PL spectra were obtained from a PL spectrometer (Edinburgh Instruments, FLS 900) excited with a picosecond pulsed-diode laser (EPL-640) with the wavelength of 638.2 nm while cooling down with liquid helium. Current density–voltage (*J*–*V*) curve of cells was collected on a Keithley 2400 Source Meter under AM1.5 G illumination (100 mW cm^{-2}) from a solar simulator (Zolix SS160A). The intensity of the light source was calibrated by a standard silicon solar cell certified by NIMC (National Institute of Metrology, China). The reference cell was certified by NIMC. Spectral mismatch was calculated by NIMC. The measurement was performed at room temperature (25 °C) in air with a scanning rate of 130 mV s^{-1} from −0.05 V to 0.6 V with a step of 6.5 mV and without dwell time. External quantum efficiency (EQE) was obtained from an Enlitech QE-R test system with a xenon lamp and a bromine tungsten lamp as light sources. TPC and TPV spectra were obtained by our lab-made setup, in which the cell was excited by a 532 nm pulse laser (Brio, 20 Hz, 4 ns) and the photovoltage decay process was recorded by a digital oscilloscope (Tektronix, DPO 7104). The *C*–*V* profiles, driven-level capacity profiles (DLCP) and temperature-dependent open-circuit voltage variation were measured on an electrochemical workstation (Princeton, Versa STAT).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated or analysed in this study are included in the published article, its Supplementary Information and Source data files. Source data are provided with this paper.

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Author contributions

The manuscript was written through contribution of all authors, and all authors have approved the final version of the manuscript. Q.M., H.X., D.L.: supervision, discussion, writing-review and editing. J.Z. and X.X.: experiments, simulation, characterization, writing-original draft. H.W.: discussion, Se partial pressure and IPCE measurement. J.W., L.L., K.Y., Y.G.: data analyses and discussion. J.S.: discussion, m-TPC/TPV analyses. Y.L.: data analyses.

Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	The total area of kesterite solar cells used for certification is 0.2668 cm2 (certified by National PV Industry Measurement and Testing Center, NPVM). The information can be found in Supplementary methods.
Method used to determine the device area	☒ Yes ☐ No	The area of certified cells are measured by NPVM.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☐ Yes ☒ No	For kesterite solar cell, no difference in J-V for forward and backward scan. Hysteresis is not oberved.
Voltage scan conditions *For instance: scan direction, speed, dwell times*	☒ Yes ☐ No	reverse scanning direction, 130 mV s-1, no dwell time, all are provided in the Characterization part in Methods.
Test environment *For instance: characterization temperature, in air or in glove box*	☒ Yes ☐ No	in air, provided in the Characterization part in Methods.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning was used in this work.
Stability of the J-V characteristic *Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.*	☐ Yes ☒ No	Kesterite solar cells are very stable. Generally, there is no need about time evolution of the maximum power point or with the photocurrent at maximum power point.and cell parameters for the fresh device and being stored in air for 3 months, see Supplementary Table 3.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	No hysteresis and other unusual behavior were observed.
Related experimental data	☐ Yes ☒ No	No hysteresis and other unusual behavior were observed.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	EQE was provided in Figure 5b.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The integrated Jsc values from EQE were basically comparable to experimental Jsc values from J-V measurements, showing less than 3% difference.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	No used in tandem solar cells.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	The intensity of the light source was calibrated by a standard silicon solar cell certified by NIMC (National Institute of Metrology, China).

Confirmation that the reference cell was calibrated and certified

☒ Yes ☐ No

The reference cell was certified by NIMC.

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes ☐ No

Spectral mismatch was calculated by NIMC.

6. Mask/aperture

Size of the mask/aperture used during testing

☐ Yes ☒ No

All J-V curves were measured without mask.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes ☒ No

All J-V curves were measured without mask.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes ☐ No

The efficiency of kesterite solar cell was certified by National PV Industry Measurement and Testing Center, NPVM (Fujian, China)

A copy of any certificate(s)
Provide in Supplementary Information

☒ Yes ☐ No

Provided in Supplementary Figure 25

8. Statistics

Number of solar cells tested

☒ Yes ☐ No

24 cells for statistical results, provided in the manuscript

Statistical analysis of the device performance

☒ Yes ☐ No

has already done in the manuscript

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions
For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes ☐ No

Supplementary Table 3.
Long-term stability, stored in ambient environment without encapsulation.